

Re-analysis Summary Report for Peer-evaluation

Paper ID: McDevitt_JournPoliEco_2014_yQeR

Re-analyst's ID: ww7ab

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: <https://forms.gle/EAREaC6JjN2PWFfR6>

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:

The average plumbing firm whose name begins with A or a number receives... more service complaints than other firms ...(p. 909.)

The corresponding article

Here you can find the original article: https://osf.io/6fusy/?view_only=c874fcf097644e82a320679d4641163a and the original data: https://osf.io/z7mfu/?view_only=277b7d18bb24436eb5cfca69a07eb07a

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:

The paper claims that plumbers who start with the letter A receive more complaints than others. I described the data and checked the distribution of the number of claims. Given its skewed distribution, I decided to model it as a count variable and dichotomise it for an alternative check. Firstly, I checked bivariate association between having a name starting with an A and number of complaints received with simple t-tests (continuous complaints measure) and chi-squared tests (binary complaints measure). To account for further control variables, I ran a number of multivariate analysis: I used logistic regressions with the dichotomised variable and negative binomial and zero-inflated poisson regressions with the continuous variable (poisson regressions did not seem appropriate due to overdispersion). I calculated marginal effects at the means for better interpretation of the results. All results support the hypothesis; I find positive effects of having a name starting with A and the number of complaints received. This pattern emerges both when looking at bivariate or multivariate associations, and is stable across different specifications.

The analyst's conclusion: The average plumbing firm whose name begins with A receives more service complaints than other firms.

The analyst's categorization of the result:

The results show evidence for the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:

Use only the data of Illinois Plumbing Firms in your model. Do not restrict your sample to those firms that serve the metro Chicago area. Do not include the reviews received via yelp.com, Angie's List, and Consumer's Checkbook into your analysis. You should use the overall complaints number instead of complaints per employee.

Here, the analyst reported the following steps and results of the analysis:

The paper claims that plumbers who start with the letter A receive more complaints than others. To satisfy the instructions of task 2, I looked at the full sample and did not undertake any pre-processing steps. I described the data and checked the distribution of the number of claims. Given its skewed distribution, I decided to model it as a count variable and use a negative binomial regression and use the dummy variable indicating whether the firm starts with the letter A as only predictor variable. Its z-statistic allows me to support the claim: I find a significant effect with $z=6.61$.

Analysis language/software/code (optional to check):

Task 1: STATA

Task 2: Stata

Analysis codes: <https://osf.io/xmzh2/>